miniRUEDI Symposium 18–20 Nov 2025

Eawag, Überlandstrasse 133, 8600 Dübendorf, Switzerland

Tentative/Draft Program

Each presentation:

20 min talk

5 min questions and discussion

	Tuesday	Wednesday	Thursday
9:00–9:30		*Coffee & Snacks*	
9:30–10:40	*Welcome* Bekaert Musy	Bugher / Harvey Schilling / van Rooyen Welham	Mace Tomonaga Marion
		Break with Coffee & Snacks	
11:10–12:00	Cateland West	Seltzer Blanc	Burghard Lefeuvre
		Lunch Break	
14:00–15:10	Giroud Lightfoot Selvadurai	Helbing Peel Brennwald	Gumindoga Moser Röggla Aardema
		Break with Coffee & Snacks	
15:40–16:30	Coppens Montoya	Arruda Alshareef	*End*
		Group Dinner	

Photosynthesis and productivity rates in the Eastern Tropical Pacific Ocean over the 2023–2025 El Niño-Southern Oscillation cycle

Hedy M. Aardema[1,2], Hans A. Slagter[1], Lena Heins[1], Antonis Dragoneas[1], Gerald H. Haug[1,2], Ralf Schiebel[1]

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(2) Department of Earth and Planetary Sciences, ETH Zurich, Switzerland

The El Niño-Southern Oscillation (ENSO) is a natural climate variability associated with strong fluctuations in the ocean environment that impact biota and productivity. This provides suitable conditions for testing the response of ecosystems to rapid climate change. We conducted five expeditions in the Eastern Tropical Pacific to study the impact of this climate variability on phytoplankton photophysiology and productivity. On the S/Y Eugen Seibold the underway measurement set-up includes a LabSTAF active fluorometer (Chelsea, UK) and a miniRUEDI mass spectrometer (Gasometrix, Switzerland) to estimate surface ocean photophysiology and dissolved oxygen-to-argon ratios, respectively. These two methods allow for the estimation of the two end members of productivity rates in the ocean, photosystem II electron transport rates and net community productivity rates. Our preliminary results show that during the El Niño phase phytoplankton standing stocks and volumetric photosynthesis rates are strongly declining. The net productivity rates of the communities do not reflect these declines, but instead are spatially and seasonally decoupled, which could modulate the efficiency of organic matter export to the deep ocean.

Evaluating the effects of multiple variable on the dissolution of CO_2

Mohammed Alshareef, Darren Hillegonds, Chris Ballentine

University of Oxford

In a controlled tube experiment, we simulated the injection of CO_2 through a vertical water column to investigate how bubble size and injection flux influence CO_2 dissolution dynamics. Continuous gas composition data were collected from the headspace using a miniRUEDI, allowing real-time monitoring of CO_2 concentrations. Analysis of the time-series data reveals distinct signatures associated with each variable, providing insights into the relationship between bubble dynamics, gas dissolution efficiency, and flux behavior.

Noble gas concentrations in a brazilian semiarid aquifer

Natália de Souza Arruda, Hannah Eckert, Bertram Graf von Reventlow, Edith Engelhardt, Jürgen Sültenfuß, Didier Gastmans, Werner Aeschbach

Heidelberg Universitat

He3/He4 as tracers to understand the groundwater flow model; Paleoclimate evidence; Excess Air; Sedimentary aquifer system.

ABSTRACT TITLE

David Bekaert

CRPG

ABSTRACT TEXT. ABSTRACT TEXT. ABSTRACT TEXT. ABSTRACT TEXT. ABSTRACT TEXT. ABSTRACT TEXT. ABSTRACT TEXT.

Impact river work entangled with gas data

Théo Blanc, Friederike Currle, Morgan Peel ...etc... Brunner Roki Schilling

Universiy of Neuchatel / Eawag

Entangle the travel time and mixing ratios from freshly infiltrating river water into groundwater. Use environmental gas tracers miniRuedi and Rad 7. Binary mixing models coupled with excess air model. The travel times change drastically because of riverwork and high discharge event.

Tracking CO_2 injection to an aquifer for CCS in Iceland

Matthias S. Brennwald, Jonas Simon Junker, Chuan Wang, Rolf Kipfer, Anne Obermann, Martin Voigt, and Alba Zappone

Eawag

Mineral sequestration of CO_2 in basalt aquifers is a promising pathway for permanent carbon storage. Within the DemoUpStorage R&D project, CO_2 was dissolved in saline groundwater and co-injected with helium as an inert tracer into a deep basalt aquifer at a pilot injection site near Helguvik, Iceland. We tested a combination of two novel geochemical and geophysical tools to monitor the subsurface CO_2 dynamics. Real-time, on-site monitoring of dissolved gases was conducted over one year using a gas-equilibrium membrane-inlet mass spectrometer (GE-MIMS). Electrical resistivity tomography (ERT) was used to track changes in aquifer resistivity before and after injection. GE-MIMS measurements revealed a clear increase in He concentrations downstream of the injection point, confirming fluid transport. However, no corresponding CO_2 increase was observed, indicating substantial CO_2 retardation relative to the fluid transport. No anomalies in He or CO_2 were detected in the overlying freshwater aquifer, indicating minimal upward migration. ERT data showed localized resistivity decreases suggesting basalt dissolution near the injection borehole. Together, GE-MIMS and ERT provided complementary, cost-effective insights into CO_2 transport, reaction, and storage processes in the basalt aquifer. These tools enhance monitoring and assessment capabilities, supporting the development of safe and effective geological CO_2 storage.

Estimating ebullition from wetlands using porewater gas concentrations

Nicolette Bugher and Charles Harvey

Civil and Environmental Engineering, MIT, USA

Methane is transported from wetlands by a variety of processes: ebullition (exsolvation of bubbles that rise to the surface), flushing (advection by pore water discharging into streams), and diffusion (both into plant roots and out through the surface). The competition between these processes leads to different porewater concentrations of dissolved gases. We hope to combine models of these competing processes with measured porewater gas concentrations in different wetlands to estimate methane fluxes from these wetlands. In particularly, we hope to estimate ebullition of methane from measurements of noble gasses in several wetlands in the eastern US. We also plan to use measured porewater gas concentrations to test hypotheses about why carbon dioxide concentrations are much lower in tropical peatlands than northern peatland.

Establishment of a new Groundwater Dating Laboratory at TU Dresden

Diana Burghardt, Paul Fremdling

TU Dresden, Institut für Grundwasserwirtschaft

Groundwater is the most important source of drinking and industrial water worldwide. Groundwater models and average residence times of groundwater are needed to assess usable groundwater reserves and their sustainable management. The only reliable method for dating sustainably usable groundwater is the analysis of helium isotopes dissolved in the water. Funding from the EFRE-InfraProNet 2021-2027 support programme and the Free State of Saxony will be used to establish a laboratory for groundwater dating using helium isotope analysis at the Institute for Groundwater Management at TU Dresden from 2026 to 2027. In addition to the funds for the system of all devices, personnel funds are also available. The objectives of the funding are (i) the technological establishment of this laboratory, (ii) the verification of helium isotope analysis of the new system in comparison to that of the helis laboratory at the University of Bremen, (iii) the validation of helium isotope analysis of the new system on representative groundwater samples in Saxony, and (iv) the development of a web map of sustainably usable groundwater reserves in Saxony.

Origin of fluid emissions along crustal to lithospheric faults in an extensional setting, Betic Cordillera, Spain

Bérénice Cateland, Anne Battani, Matthais Brennwald, Benjamin Lefeuvre, Nicolas E. Beaudoin, Frédéric Mouthereau, Rolando Burga, Jose Benavente, Magali Pujol

Université de Pau et des pays de l'Adour

The Betic Cordillera (SE Spain) has experienced a complex geodynamic history involving crustal thinning related to slab retreat. This deep lithospheric dynamic was accompanied by alkaline to calc-alkaline volcanism and the exhumation of metamorphic domes. These processes feed an active fluid system, the respective contributions of which remain poorly constrained. However, understanding and quantifying these deep fluid circulations has major implications for geothermal exploration and natural gas migration. Several crustal to lithospheric-scale faults in the Internal Betics have accommodated most of the deformation. These active structures (Lorca earthquake, Mw 5.2) are associated with significant hydrothermalism, as indicated by water temperatures reaching up to 65 °C, and may act as preferential pathways for fluid and heat migration. We sampled fluids (water and gas when bubbles were present) from 14 thermal springs with temperatures ranging from 15 to 65 °C. Gas species were measured in the field using a portable mass spectrometer (MiniRUEDI) along these strike-slip faults. The Cádiz-Alicante Fault releases fluids rich in N2 (86–96%), with CO2 contents between 1 and 5%, and O2 between 0.5 and 10%. Along the Alpujarras Fault, CO2 dominates (63–91%) with lower amounts of N2 (5–28%). The lithospheric Carboneras Fault emits fluids rich in N2 (91%), with low proportions of O2 (3%) and CO2 (3%). Alhama de Murcia, the most active fault of the system, mainly releases CO2 (52%) and N2 (43%). This result helps us in discussing fluid origin in this extensive context.

Vermifiltration gas emissions and mass balances

Kayla Coppens

Eawag / University of Geneva

The miniRUEDI was used with a static flux chamber to quantify CO_2 and CH_4 emissions from 6 laboratory-scale vermifilters treating blackwater and brownwater at varying hydraulic loading rates. The gas emissions were further used to explore the impact of wastewater type and hydraulic loading on nitrogen and carbon mass balances.

Continuous miniRUEDI measurements reveal gas-seismic interactions at Lavey-les-Bains

Sébastien Giroud, Matthias Brennwald, Rolf Kipfer

Eawag

The link between dissolved gas concentrations in geological fluids and seismic activity remains debated. Reported correlations between gas composition changes and earthquakes are typically anecdotal and event-specific, while systematic, long-term datasets are rare [1]. To overcome this limitation, we deployed a portable gas equilibrium membrane-inlet mass spectrometer (miniRUEDI; [2]) at the geothermal springs of Lavey-les-Bains (50 °C to 65 °C), located in a seismically active region of Switzerland. The instrument continuously recorded dissolved gas concentrations (He, Ar, Kr, N_2, O_2, H_2, CH_4, and CO_2) over more than three years, providing one of the longest high-frequency datasets available. We use these time series for a critical assessment of the temporal correlation between seismic events and gas evolution in geothermal fluids.

[1] Toutain and Baubron (1999), Tectonophysics, 304, 1–27

[2] Brennwald et al. (2016), ES&T, 50, 13455–13463

Zimbabwe-Headwater Catchment Response to Extreme Rainfall Activities-Meeting on sampling of hydrological extreme events

Webster Gumindoga

University of Zimbabwe

Extreme rainfall events are becoming increasingly frequent across Zimbabwe, yet their effects on headwater catchment hydrology remain poorly quantified. Understanding rainfall–runoff–groundwater interactions under such events is essential for water resources planning, particularly in convective rainfall-dominated systems. This study assessed the hydrological and isotopic response of three representative headwater catchments—Domboshava (Mazowe), Marondera (Manyame), and the University of Zimbabwe (Gwebi)—during the 2023/2024 wet season. Stable isotopes of oxygen (δ^{18}O) and hydrogen (δ^{2}H) were analysed in rainfall, surface water, and groundwater, while Radon-222 (^{222}Rn) was used as a tracer of subsurface flow dynamics. Sampling was conducted at event and intra-event scales, with rainfall collected up to three times daily and groundwater monitored through logging wells. Early-season rainfall at the University of Zimbabwe site exhibited isotopic enrichment (δ^{18}O = 3.06‰; δ^{2}H = 17.48‰) and low d-excess (-7‰), indicating subcloud evaporation and limited recharge potential. Intra-event sampling revealed "Λ-shaped" isotopic profiles, reflecting the temporal variability within convective storm systems. Distinct isotopic signatures were observed among rainfall, surface water, and groundwater, underscoring episodic recharge patterns. Variability in ^{222}Rn concentrations indicated contrasting subsurface flow paths and groundwater residence times across sites. The study highlights the heterogeneity of recharge processes in Zimbabwean headwater catchments and the influence of convective rainfall on water partitioning. Data limitations due to manual sampling and equipment gaps remain key challenges. Future work will focus on improved monitoring networks, recharge zone mapping, and long-term isotope-hydrology integration for water resource resilience under extreme rainfall conditions. Keywords: Stable isotopes, Convective rainfall, Groundwater recharge, Headwater catchment, Radon-222, Zimbabwe.

Noble gases – suitable tracers to optimize the productivity of a managed aquifer system?

Jakob Helbing, Marie-Sophie Maier

Zurich Water Supply, Switzerland

The Hardhof groundwater field is essential for a secure drinking water supply in the city of Zurich. The city's urban infrastructure surrounds the groundwater field and thus limits its local extension. To increase the natural groundwater pumping capacity and guarantee safe drinking water supply at any time, the groundwater field is artificially managed. Riverbank-filtered Limmat water is pumped across the field and injected into the ground along the most vulnerable side towards heavy traffic infrastructure. To determine the necessary amount of water injected into the ground, a sophisticat-ed real-time hydrological numerical model is employed. For an optimal performance, this model needs to be calibrated and evaluated. One possibility to do so is identifying real flow paths with tracer experiments and comparing these results with the output of the numerical model. In a groundwater protection zone however, nonhazardous tracers are vital. The groundwater field Hardhof offers a possibility to test the assumption, whether noble gases can be used as tracers in large scale managed aquifer recharge (MAR) systems. Approximately 100 boreholes within the groundwater field can be used both as injection points and measurement locations for noble-gas tracers. Therefore, different setups of distance and spread can be well tested. In our presentation, we will discuss experimental design and first estimations of this project.

Installation of a noble gas laboratory in Pau

Benjamin Lefeuvre, Anne Battani

LFCR - UPPA

oble gases (He, Ne, Ar, Kr, Xe) are excellent natural tracers for studying geological processes, as their fully filled outer electron shells make them chemically inert. A new noble gas laboratory has been established in Pau through joint funding from TotalEnergies and the University of Pau, dedicated to the characterization of these gases in natural samples. Two mass spectrometers (an Argus VI and a Helix SFT from Thermo Fisher Scientific) have been acquired and installed. Prior to sample introduction into these instruments, a custom purification and separation line was developed. This line enables the removal of reactive gases through a series of physico-chemical traps, such as cold fingers filled with activated charcoal, Zr–Al getters, and titanium sublimation pumps (TSPs). Once purified, the remaining noble gases are cryogenically separated and introduced into the mass spectrometers for isotopic analysis.

In addition, a quadrupole mass spectrometer (QMG 250, Pfeiffer Vacuum) has been installed to monitor the purification process and to analyze major gaseous components. This instrument is similar to the MiniRuedi mass spectrometer and is equipped with two detectors: a Faraday cup and an electron multiplier allowing the quantification of gases at both high and low concentrations.

Dissolved Gas Monitoring at the Bedretto Underground Laboratory: Responses to Seismic Stimulation

Alexandra Lightfoot, Matthias Brennwald, Valentin Gischig, Alba Zappone, Rolf Kipfer, Stefan Wiemer

SED ETH Zürich

Gas Monitoring at the Bedretto Underground Laboratory: Responses to Seismic Stimulation In hydraulic stimulation experiments, such as those used in seismic and geothermal energy studies, monitoring dissolved reactive and noble gases in groundwater can provide key insights into fluid transport and redistribution. While tracers added to injection water reveal specific flow paths, dissolved gas monitoring in general captures a broader picture of how fluids are mobilised or redirected under pressure. At the Bedretto Underground Laboratory for Geosciences and Geoenergy (BULGG), we tested the geochemical response during an experiment designed to trigger a controlled magnitude-zero earthquake. High-pressure water (up to 20 MPa) was injected into borehole ST1, with krypton added as a tracer. A nearby borehole (ST2), 44 m away and discharging continuously, was equipped with a miniRuedi to measure dissolved He, Ar, Kr, N2, O2, and CO2 in real time, complemented by parallel radon monitoring. This setup enabled us to track both immediate gas responses to pressure and delayed tracer migration. Dissolved gas signals at ST2 revealed a clear response once injection began. N2 and Ar increased in terms of partial pressure, while He decreased, showing that injection resulted in mobilised or redirected younger, less radiogenic fluids toward ST2 well before the Kr tracer arrived. Radon activity increased only after tracer breakthrough, consistent with fracture opening and fluid movement under stress. These findings highlight dissolved gas monitoring as a sensitive tool for probing the interplay between stress, fractures, and fluids, particularly in the context of hydraulic stimulation.

Need to figure out a title

Wil Mace

Universtiy of Utah

Membrane Inlet Mass Spectrometry (MiniRuedi) is a powerful tool for measuring dissolved gases in real time directly in the field. Recent studies have shown that changes in gas composition detected with this technique can be linked to activity in hydrothermal and tectonic systems. While measuring $^3He/^4He$ ratios could also provide valuable information, the MiniRuedi cannot measure these directly, and collecting suitable samples is not easy. However, since the MiniRuedi can track several dissolved gases at once–and because changes in these gases have been connected to seismic activity–it can be used to trigger the collection of gas samples. These samples can then be sent to a noble gas laboratory for precise measurement of $^3He/^4He$ ratios. Here, we introduce a prototype sampling box designed to collect and preserve gas samples for high-precision analysis.

New tool for in-situ gas measurements in low flow porous media

C. Marion, C. Ebi, S. Bloem, M.S. Brennwald, R. Kipfer

Eawag

Dummy dummy to be improved and checked by co-authors: Gases, including noble gases, are powerful tracers in environmental science, widely used to study groundwater flow dynamics and surface–subsurface water interactions (Brennwald et al. 2022, Blanc et al. 2024, Schilling 2017, Schilling 2021, Peel 2022 & 2023). However, most applications focus on high-flow systems (L/min to L/s), while in-situ gas measurements in low-flow porous media (L/h), such as soil unsaturated and saturated zones or plant tissues, remain technically challenging—despite their critical role in the global hydrological cycle. We present a robust, field-deployable gas probe for long-term, in-situ gas monitoring in low-permeability or low-flow environments. The device consists of a gas-tight, 3D-printed body with an integrated pressure sensor and gas-permeable membranes that allow diffusion of target gases into the probe's headspace for subsequent sampling and analysis. Originally developed for in-vivo gas monitoring in tree xylem (Marion et al., 2024), this method enables non-destructive, high-resolution measurements applicable to a variety of porous media, including soil, sediment, rock, and biological tissue.

NamCo / Tibet (preliminary)

Paul Moser Röggla, etc.

Eawag

NamCo lake, gas data (Preliminary)

methane emissions from onsite sanitation containments

Natalia Andrea Montoya Arévalo

Eawag

a short presentation about the use of the miniRuedi to measure methane emissions from onsite sanitation containments.

Field Deployment of a miniRUEDI for Long-term Gas Monitoring in the Mt. Fuji Volcanic Aquifer System

Stephanie L. Musy[1], Yama Tomonaga[1,2], Friederike Currle[1], Naoto Takahata[3], Yuji Sano[4], Oliver S. Schilling[1,5]

(1) Hydrogeology, Department of Environmental Sciences, University of Basel, Switzerland
(2) Entracers GmbH, Dübendorf, Switzerland
(3) Atmosphere and Ocean Research Institute, University of Tokyo, Kashiwa Campus, Chiba, Japan
(4) Marine Core Research Institute, Kochi University, Monobe Campus, Kochi, Japan
(5) Eawag, Swiss Federal Institute of Aquatic Science and Technology, Dübendorf, Switzerland

A portable quadrupole mass spectrometer (miniRUEDI) was installed at a drinking water well intersecting the Fuji–Kawa–Kako fault zone — one of Japan's most active fault systems — within the Mt. Fuji catchment. The system continuously monitors noble and reactive gases in both ambient air and groundwater using the gas-equilibrium membrane inlet mass spectrometry (GE-MIMS) approach. This long-term deployment, ongoing since 2023, provides new insights into hydrological and geochemical dynamics within Mt. Fuji's volcanic aquifer system. Distinct variations in CO_2, O_2, and ^4He concentrations were observed following major precipitation events, indicating the infiltration of newly recharged meteoric water. Additional transient gas anomalies coincided with regional seismic activity, suggesting a potential link between crustal stress changes and subsurface gas transport. Despite maintenance requirements due to water condensation and membrane module clogging, the system demonstrated stable, long-term operation. Overall, the results highlight the potential of continuous gas monitoring for resolving coupled hydrological, geochemical, and tectonic processes in active volcanic environments.

Multi-Tracer Insights into Urban Catchment Dynamics and Wetland Support

Welham, A., van Rooyen, J., Watson, A., Chow, R., Roychoudhury, A.

Eawag

This contribution presents a multi-tracer study of an urbanized catchment in South Africa, combining isotopes (δ^{18}O, δ^{2}H), tritium, major ions, and dissolved natural gases to resolve interactions between rainwater, rivers, groundwater, and surface water transfer schemes. The results highlight how event-driven recharge, longer-residence groundwater, and managed surface flows jointly shape estuarine wetland resilience under climatic and anthropogenic pressures. (Will send the full abstract later today)

Seasonality of Excess Air Signatures Under Varying Flow Conditions in a Sub-Alpine Fractured Bedrock Aquifer

Christoph E. West, Nicholas E. Thiros, Werner Aeschbach, W. Payton Gardner

Institute of Environmental Physics Heidelberg

We report on a tracer study to constrain groundwater dynamics in an alpine groundwater system near Crested Butte (Colorado, USA), dominated by snowmelt patterns. The hydrograph in this area, underlain by fractured shale, shows two flow periods: a baseflow period (BP) lasting from late summer to early spring, where no recharge occurs and the water level is 3-4 m below land surface; and a recharge period (RP) starting in late spring caused by snowmelt, leading to a rapid rise in water level by 1.5 m, which then declines until late summer. Dissolved noble gases were sampled at different times from a 10 m deep bedrock well at the bottom of a hillslope transect close to the East River. We observe different DeltaNe signatures during both flow periods: positive DeltaNe during RP and negative DeltaNe during BP. Our preliminary results suggest a link to (1) the hydrostatic pressure and (2) prevalent CO_2, N_2, and CH_4 subsurface production caused by bedrock weathering. Lower hydrostatic pressure during BP results in exsolution of oversaturated gases, which may be enhanced by continuous subsurface gas production. At higher hydrostatic pressure during RP, the positive DeltaNe from recharge may prevail. Our results highlight the complexity and seasonality of groundwater dynamics in snowmelt driven mountainous fractured bedrock aquifers, furthered by additional processes like subsurface gas production.

Noble gases as artificial tracers for routine hydrogeological investigations

Morgan PEEL, Kai SOLANKI, Daniel HUNKELER, Philip BRUNNER, Rolf KIPFER

Eawag

Noble gases are increasingly adopted as artificial tracers for hydro(geo)logical studies, thanks to advances in portable, high-resolution dissolved gas measurement technologies [1-3]. Their inert, non-toxic, and invisible nature makes them ideal for sensitive environments, such as rivers and drinking water catchment areas, where maintaining water quality and appearance is essential. The application of gases in aqueous environments poses unique challenges compared to traditional tracers, as potential degassing needs to be accounted for, and ideally avoided, during tracer injection. We present a straightforward method for preparing and injecting tracer solutions with accurately controlled dissolved gas concentrations, whilst avoiding tracer loss. A two-step degassing-pure gas saturation approach enables rapid preparation of high-concentration tracer solutions using readily available materials. We applied this method in a riverbank filtration setting to perform well-to-well tracer tests using helium-4 (He-4) and krypton-84 (Kr-84). Known quantities of both tracers were injected into observation wells upstream of a pumping well. A portable mass spectrometer ("miniRUEDI" GE-MIMS [3]) was used for continuous monitoring in the pumping well. Breakthrough curves of He-4 and Kr-84 enabled precise determination of groundwater flow velocities, travel times, and tracer recovery. These findings illustrate how noble gases complement traditional tracer methods in hydrogeological investigations. The method is adaptable to other gases (e.g., Ne, Kr, Xe, light hydrocarbons), significantly expanding the range of tracers available in groundwater management contexts. 1 : Brennwald et al., 2022, Front. Water (4) 2 : Blanc et al., 2024, Wat. Res. (254) 3 : Brennwald et al., 2016, Env. Sci. Techn. (50)

Investigating gasgeochemical production during the preparatory phase of dynamic brittle failure in Rotondo granite in the laboratory

Paul Selvadurai, Claudio Madonna, Clement Roques, Hanna Ventura, Matthias Brennwald, Rolf Kipfler, Benoit, Valley, Alexandra Lightfoot, Laurent Marguet

ETH Zurich

Laboratory geochemistry studies enhance our understanding of earthquake precursors. Controlled experimental conditions allow researchers to observe how microcracks, fluid flow, and mineral reactions influence geochemical signals, including gas emissions and isotope distributions. These experiments can identify specific geochemical changes, such as the release of radon or noble gases, associated with rock fracturing. Such insights are crucial for developing indicators of large events while calibrating geochemical signals, establishing links between these signals, stress states, and damage levels within rocks.

This study investigates the use of helium, argon, and CO2 as real-time geochemical tracers for brittle deformation in crystalline rocks, focusing on Rotondo granite from the Bedretto Laboratory. These gases, stored in mineral lattices or fluid inclusions, are released during microstructural damage; however, their behavior during progressive rock failure remains poorly understood. Uniaxial compression experiments monitored gas variability with a miniRuedi quadrupole mass spectrometer, complemented by spatial strain measurements using Distributed Fibre Optic Strain (DFOS) sensing, acoustic emission (AE) monitoring, and post-mortem X-ray computed tomography (XRCT) of fracture networks. The experimental design aims to identify distinct deformation stages, from crack initiation to macroscopic failure, and link these to transient noble gas signals. Preliminary results reveal stage-specific fluctuations in helium and CO2, supporting the hypothesis that gas emissions reflect structural damage evolution. This research advances the integration of mechanical and geochemical monitoring in rock mechanics, with important implications for underground stability assessment and early warning systems in geotechnical and seismic environments.

Year-Long miniRUEDI Gas Time Series Reveal Shutdown-Driven Excess Air and Water-Quality Impacts

Jared van Rooyen, Yama Tomonaga, Matthias S. Brennwald, Rolf Kipfer, Oliver S. Schilling

Eawag

Managed aquifer recharge (MAR) operations can unintentionally modulate subsurface gas dynamics, with consequences for redox conditions and groundwater quality. In this study, we deploy a portable mass spectrometer (miniRUEDI) in an observation well at the Hardwald MAR scheme near Basel, Switzerland, and acquire continuous noble- and reactive-gas records (He, Ar, Kr, N_2, O_2, CO_2) over a one-year period. The instrument enabled unattended, on-site measurements with percent-level precision and high temporal resolution, well-suited to capture transient events. During the study, MAR operations ceased eight times for periods ranging from hours to several days. Each cease produced a potential characteristic multiphase gas response: (i) rapid shifts in dissolved gas ratios consistent with bubble entrapment and dissolution upon re-wetting, i.e., formation of "excess air"; (ii) step-changes in O_2 and N_2 that propagated on operational time scales; and (iii) knock-on effects on redox-sensitive species (CO_2, Ch_4) indicative of short-term biogeochemical reorganization. By combining noble-gas diagnostics of excess air with reactive-gas trends, we separate physical recharge effects from biogeochemical responses and quantify how shutdowns alter effective recharge signatures and residence-time proxies. These findings highlight portable mass spectrometry as a process-control tool for MAR, linking operational decisions to subsurface gas dynamics and water-quality outcomes in near-real time. Our results generalize established concepts of excess-air formation in porous media to the operational rhythms of MAR and provide actionable guidance for minimizing adverse water-quality swings during start/stop cycles.

Continuous in situ-based gas measurements for volcanic gas monitoring, developing a membrane degasser, and profiling gases in a stratified pond

Alan Seltzer

Woods Hole Oceanographic Institution, Department of Marine Chemistry and Geochemistry and University College Dublin, School of Earth Sciences

I will share a brief and informal overview of different applications of in-situ miniRUEDI measurements by our group over recent years and present a few avenues for potential future developments and collaboration. These include (i) pilot results for monitoring of CO_2-dominated volcanic gas emissions in bubbling springs, (ii) testing of a prototype membrane-based degasser, and (iii) depth profiling of CH_4 and other dissolved gases in a highly stratified brackish pond.

For part (i): Previous work by our group and others has shown that noble gases in surface CO_2 degassing sites (e.g., hot and cold springs) show high variability both spatially (e.g., two bubbling plumes within a meter might vary widely in noble gas composition) and temporally (e.g., repeat sampling of the same site has revealed considerable variation over time). In these systems, where Ne, Ar, Kr, and Xe are largely dominated by a groundwater (i.e., atmospheric) component that has degassed into an upwelling CO_2 gas phase, noble gas elemental ratios typically vary between air-like and air-saturated-water-like values, while recent heavy noble gas isotope measurements have shown extensive kinetic fractionation associated with intermediate elemental ratios. Continuous monitoring of noble gases in such systems may shed light on the mechanisms of water-gas interaction, which in turn might inform larger-scale relationships between volcanic and hydrogeologic systems in areas prone to eruption.

For part (ii): Together with a group of undergraduate student fellows, we developed a simple single-membrane degasser for the collection of sufficiently large quantities of gas to analyze the isotopic composition of rare gas species (e.g., Hg) in water. To evaluate the performance of a single-membrane degasser with respect to individual gas species, with the goal of upscaling such a system to achieve quantitative degassing for higher solubility rare gases, we carried out tests in WHOI's AVAST center in a saltwater tank in which a miniRUEDI GE-membrane was placed downstream of the degasser to constrain gas-specific extraction efficiencies to tune a model that could be extended to other gases.

For part (iii): To investigate physical and biogeochemical dynamics of a brackish, meromictic pond in Falmouth, MA, we carried out continuous dissolved gas measurements using a miniRUEDI aboard a small dinghy. Using a submersible pump that we gradually lowered to different depths, we analyzed noble gases, N_2, O_2, and CH_4, producing a data set that may be useful in ultimately constraining the flux of CH_4 through the pond, while also providing an interactive educational opportunity in both mass spectrometry and environmental biogeochemistry to a group of undergraduate students.

A (very) portable Gas Equilibrium Membrane Sampling System (GEMSS)

Yama Tomonaga, Oliver S. Schilling

Eawag / University of Basel

The miniRUEDI has proved to be a valuable tool to monitor continuously and in real time the gas dynamics in environmental systems. While mid- and long-term observations imply generally fixed installations with a relatively constrained effort in terms of logistics, the spatial screening of the gas composition in surface waters and groundwater requires the transport of miniRUEDI with the respective GE-MIMS (and a power source, e.g. a generator) to the different sampling locations. This can be sometimes problematic, as the instrument weight of about 30 kg (not including pumps, generator ...) makes it ”portable” only to a certain extent. To address this limitation, and to simultaneously improve on the long standing limitation of samples taken in copper tubes not being prescreenable on the miniRUEDI prior to time-consuming noble gas isotope analyses, we developed a small portable Gas Equilibrium Sampling System (GEMSS aka ”James”) that can be taken to the field easily and allows for a sufficient amount of gas to be sampled for multiple analyses on a single sample. The system uses the same concept of gas equilibrium of the conventional GE-MIMS. However, instead of functioning as an inlet for miniRUEDI measurements, James allows acquisition of gas samples for subsequent analyses in reusable containers. James is relatively light and fits in a backpack, it not only allows replicate measurements on a miniRUEDI, but the respective samples can further be used also for other types of analyses (e.g., conventional noble gas isotope analyses on magnetic sector mass spectrometers).